FIN1209 Technical Analysis in Investment

Chapter 1 teaching notes

Introduction to the Art and Science of Technical Analysis

209 slides **25** checks, 50 items **31** figures **6** parts

Benjamin C. Sotelo, Institute of Accounts, Business and Finance, FEU Manila

Read this first

Pick a run plan before you open the deck

Why this exists

The deck is 195 minutes at full length and most sessions are not. Nobody has confirmed how long this one is, so choose a plan now rather than discovering the problem at the 90 minute mark. All four plans below cover the same chapter and all seven booklet objectives. They differ in how much of it you say out loud.

Any plan

RULE **If you fall behind mid-session**

Drop the next figure that only redraws a slide you have already taught. **Never a check, and never anything between Figures 1.9 and 1.15.**

The decision

MINUTES PER PART, BY PLAN

PART	SLIDES	FIGS	FULL	LONG	STANDARD	SHORT
1 Why we analyze	25	1	26	22	13	10
2 Forecasting	42	3	33	24	22	15
3 Classifications	27	3	28	23	15	12
4 Subjectivity	39	13	42	34	20	15
5 Assumptions	47	8	38	26	22	16
6 Participants	22	3	28	21	13	7
Whole chapter	202	31	195	150	105	75
Checks kept, of 25			25	25	25	19
Minutes that are checks			50	50	50	38
Minutes that are teaching			145	100	55	37

Slide counts run from the part divider to the recap, which is what the progress marker counts. The other seven slides are the openers and closers: five minutes, in every plan.

Read this

Read the last three rows first. Teaching content falls by three quarters between Full and Short. Checks fall by a quarter. That is deliberate and not negotiable: the checks are how you read the room. Cut talk faster than you cut checks.

Before you start

A live chart in a second window, because Parts 1, 4 and 5 all want one. **The board free**, because Parts 1 and 5 both want writing that stays up for the rest of the session. **Nothing else.** No handouts, no printing.

Standing instructions

How this deck is meant to be run

Which deck

Teach from the locally built deck with the artwork in it, not the committed one, which carries placeholders where the 31 figures go. `README.md` here has both build commands, and `in-class-checks.md` has every question in full.

The shape

The deck is built for a room that includes students with ADHD, so it is many small slides rather than a few dense ones. Do not slow down to fill a slide. Most teaching slides are one idea and should take under a minute.

Every slide carries its own speaker cue in the PowerPoint notes, two or three lines each, so **run the deck in Presenter View.** These notes carry what is not on the screen and not in Presenter View: the timing, the cut decision, the board work, the traps, the folds, what the department examines, and the answer letters.

Three things carry the session

1. **The progress marker.** Bottom left of every slide: `Part 3 of 6 - Classifications | 7 of 16`. Students always know where they are. If someone asks how much is left, point at it rather than answering.
2. **The checks.** Twenty five of them, two questions each, fifty items. A dark green slide with a gold CHECK chip means a question is coming. The room learns that colour within the first twenty minutes.
3. **The section boundaries.** Six parts. Each opens with what it covers and closes with a you-now-know recap. Any boundary is a clean stop if you run out of time or the room runs out of attention.

Running a check

Twenty five times today. It is the same procedure every time.

Question slide

Read Q1 aloud, then Q2. **Sixty seconds, no discussion, no phones.** Ask for a show of hands on each option before you advance. The silence is the point: it is the processing time that makes the question answerable by everyone in the room, not only by the fastest three.

Reveal slide

The reveal gives the letter, the option text and a one-line reason. **Say the reason out loud.** The letter alone teaches nothing, and immediate feedback is the half of a check that does the work.

Every check

RULE **If more than about a third of the room misses an item**

Go back one slide and reteach it. That is what the checks are for. They carry no marks and you should say so the first time, or the room will freeze.

Running a figure slide

31 of them, and 13 are in Part 4 alone.

Every figure

Name what they are looking at first. The instrument, the timeframe, the axes. Then stop talking and let the room look at it. Only then make the point, off the slide's own speaker cue. Do not improvise a second reading of the chart.

What the department actually examines

Three documents

The cut markers are not taste. They come from the department's Quiz 1, from Homework 1, and from the seven objectives printed in the course booklet.

Quiz 1

QUIZ 1 IS 20 ITEMS. ITEMS 1 TO 10 ARE CHAPTER 1.

ITEM	WHAT IT ASKS	PART	SLIDE
1	The four profitable scenarios	1	27
2	The dual function	1	15
3	Intrinsic value, with undervalued and overvalued reversed as a trap, and a CAPM line	2	34
4	The three ways to forecast, plus flow of funds	2	31
5	That technical analysis studies price and market action	2	48
6	That flow of funds means margin debt	2	67
7	Effect against cause, and the two profiles	2	65
8	Momentum, the Darvas Box, and contrarians	3	84
9	The mean reverting tool set, naming limit entry orders	3	85
10	Random walk	5	159

Items 11 to 20 are Dow Theory and belong to Chapter 2. Every row above is Core in every plan.

Homework 1

A single charting exercise worth 20 points: find live charts showing different trends, draw trendlines, interpret. That makes the price-time chart in Part 1 (slide 14) and the trendline material in Part 4 **Core, including at 75 minutes.** It is also the closing exercise of Part 4, so setting that exercise today is preparation for it.

The booklet

SEVEN OBJECTIVES, PRINTED ACROSS SLIDES 2 AND 3

	OBJECTIVE	WHERE IT LIVES
1	Understand the key concepts underlying technical analysis	Part 5, s138 to s184
2	Identify the different forms of chart analysis	Part 3, s73 to s78
3	Describe the objectives of technical analysis	Part 1, s15
4	Understand what subjectivity means in technical analysis	Part 4, s99 to s137
5	Recognize the strengths and weaknesses of technical analysis	Part 3, s88 to s91
6	Categorize market participants by style and by time in markets	Part 6, s185 to s206
7	Identify the various styles and approaches in technical analysis	Parts 3 and 6

Do not misread this

EXAMINED **Quiz 1 examines nothing from Part 4 and nothing from Part 6**

That is not permission to drop them. Objectives 4, 6 and 7 live there, review questions 3 and 4 are answered in Part 4, and Homework 1 is Part 4's closing exercise. The old "Parts 1 to 3 are a complete unit" escape hatch threw away three objectives, which is why it is gone.

Mark the deck before the session, not during it

The four run plans in detail

Short session

> **RULE** **Compress the chapter. Do not truncate it.**
>
> Parts 4, 5 and 6 carry three of the seven booklet objectives between them, so stopping at Part 3 throws away subjectivity, the assumptions and the market participants. **Do not split inside a part.**

Every plan

> **RULE** **Never cut the two learning objectives slides**
>
> Slides 2 and 3. They are the only place the students see what they are accountable for, and the seven objectives are printed across them four and three.

CORE Never cut	A booklet objective depends on it, the department's Quiz 1 tests it, or a later chapter needs it. Every part page opens with its own Core list.
REINFORCEMENT Cut when short	It teaches a core idea a second way. This is where the Long plan gets its 21 minutes back.
ENRICHMENT Cut first	Interesting, deepens the subject, nothing downstream depends on it.
FOLD Say it, do not show it	Do not show the slide, but do say the one sentence it exists for, because a check or a quiz item hangs off it. **Folding is how you lose slides without losing content.**

195 min

Full

Everything, in the order it is built.

All 25 checks. All seven objectives, at full depth. Skip nothing. Use this only if you have a genuine three and a quarter hours of teaching time after attendance and setup.

150 min

Long

Drop every Enrichment block, plus every figure that only redraws a slide you have already taught.

Why figures

The figures are what took this deck from 165 minutes to 195, so they are the cheapest 21 minutes in the chapter to hand back. Ten of the 31 survive. The slides that come off are in the table at the end of this section.

Figures

	FIGURES
Skip these 21	1.1, 1.2, 1.3, 1.4, 1.6, 1.7, 1.8, 1.16, 1.18, 1.19, 1.21, 1.25, 1.26, 1.27, 1.29, 1.30, 1.31, 1.32, 1.33, 1.34, 1.35
Keep these 10	**1.9 through 1.15 as one unbroken sequence** (slides 104 to 110), plus 1.17, 1.20 and 1.28

Checks

All 25. No exceptions. All seven objectives, fully covered.

105 min

Standard

Core only, with the folds done so no check is stranded.

The point

Everything marked Reinforcement or Enrichment comes out. Fifty of the 105 minutes are checks, which is the whole point of this plan: **it is the shortest run that still teaches the chapter properly rather than mentioning it.**

Skip

The same 21 figures as Long, plus every Reinforcement and Enrichment slide. The table below is the whole list for both plans, in deck order, so you can mark the deck once.

Pace

> **RULE** **One pass per accent line at this plan**
>
> No second example, and no show of hands on a teaching slide. **The show of hands inside a check stays, in every plan**, because that is the reading of the room you are paying for.
>
> Keep the board work in Parts 1 and 5, keep the handclap in Part 5, and keep the oil example in Part 4. Those four things cost about six minutes between them and they are the most memorable minutes in the session.

Checks

All 25. All seven objectives, fully covered. Nothing degrades.

75 min

Short

The floor. Core only, run hard, and six checks come out.

What it costs

This is the least you can teach and still say honestly that you covered Chapter 1. Skip everything Standard skips, and then these six checks: **5, 7, 16, 21, 22 and 24.** Keep 1, 2, 3, 4, 6, 8, 9, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 23 and 25. Every check you keep sits on department-examined material or on a booklet objective.

Fold harder

ON TOP OF THE STANDARD FOLDS, BY PART

PART	WHAT TO DO INSTEAD
1	Teach the four trading verbs off the board square instead of the four term slides (s21 to s24). Draw the square, say the four words, say that covering is a buy that ends a bearish position, move.
2	Run two definitions on screen, Murphy's first and Edwards and Magee on the market itself and not the goods. Say the Pring probabilities line aloud off the Murphy slide. If they remember one sentence today it is still that one.
3	Run the advantages and the disadvantages slides as a single pass. Do not take the show of hands on the two bets, just tell them neither bet is right and that this is the difficulty of the subject. The checks in this part still get theirs.
4	The sequence 1.9 to 1.15 runs exactly as written, at full speed, with the closing question. Everything around it goes to one pass. The oil example gets 60 seconds, not three minutes.
5	Teach the three forms of EMH off the ladder table on the Part 5 page rather than the three term slides (s154 to s156). Write it on the board once.
6	Seven minutes. The four contrast pairs and the discretionary pair read as a list, then the time ladder, then the method ladder, then stop.

Objective 6

EXAMINED **At Short, objective 6 degrades. Compensate, cheaply.**

Objectives one through five and seven are still covered. Part 6 gets seven minutes, so the eight categories become a list read once rather than a set taught in contrast pairs. Three compensations, all cheap: keep checks 23 and 25, because check 25's second question is the time ladder; post Figures 1.33 and 1.34 to Canvas **before** the session so the two ladders are in front of them while you read the list; and say out loud that objective 6 is examinable and is in the assigned reading.

Assign it

Reading you must assign if you run Short. Name it in class and put it in Canvas the same day. Lim, The Handbook of Technical Analysis, Chapter 1:

1. The survival instinct and its four components, and Figure 1.1 on the mechanics of profiting from a change.
2. The financial ratios, the top-down and bottom-up approaches, and all six definitions of technical analysis.
3. The second list of advantages of technical analysis.
4. Figures 1.16, 1.18, 1.19 and 1.21, and the price, time and algorithmic filters.
5. The four applied assumptions, Figures 1.25 through 1.32, and price against value.
6. All of the market participants material, including Figures 1.33, 1.34 and 1.35, and the five ways to participate in gold. **This is the objective 6 makeup and you should say so.**

Mark the deck

What comes off at Long and at Standard

Every slide that leaves the deck, in deck order, with the sentence to say when one is folded rather than dropped. Standard removes everything in this table; Long removes only the rows marked Long.

SLIDE	WHAT IT IS	OFF AT	SAY THIS INSTEAD
7	What the survival instinct contains	**Long**	The herd line, off s6, because Part 5 closes that loop.
20	What identification hands you before you risk a peso	Standard	
40	The ratios	**Long**	None of them has a time axis, and CAPM balances risk against expected return over a risk-free rate. Quiz 1 item 3 carries a CAPM line.
41, 42	Top-down and bottom-up	**Long**	Name the pair in one sentence, or check 5 loses its first question.
52	Edwards and Magee, recorded history and the word probable	**Long**	
54	Pring, the art of catching a reversal early	Standard	Say the takeaway aloud, or check 7 loses its second question.
58	Pring, people continue to make the same mistakes	**Long**	Part 5 says it again on s168.
61	What technically based timing gives you, continued	**Long**	
89	The advantages, continued	**Long**	Keep its third line: it names the self-fulfilling prophecy before s92 does.
130 to 132	The price, time and algorithmic filters	**Long**	Name all three in one minute, because check 16 asks for the time filter.
165	Price is not the same as value	Standard	
177 to 180	The four applied assumption term slides	**Long**	Read the four off the summary slide s176 as a set. Check 22 still works.
183	Where technical analysis works best	**Long**	
186	The cast of a market	**Long**	
203	Five ways to own gold	**Long**	A gold backed exchange traded fund is a derivative, because check 25 asks for exactly that.

Fifteen rows, and six of them cost you nothing but the slide. The Short plan cuts on top of all of this; the Short section above says how.

Why anybody analyzes a market

Progress marker on screen reads `Part 1 of 6 - Why we analyze`, 25 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
26 min	22 min	13 min	10 min

CHECKS	1 CA s11 2 AC s18 3 AB s25
FIGURES	1.1 s28 cut at Long
OPEN	Markets are a survival behavior before they are a financial one. Say that, then start.
CLOSE	You cannot profit from something that never moves. Leave the four-verb square on the board.
TERMS	The three motivational instincts, variable of change, price, the buy low sell high principle, the price-time chart, the dual function, identification, forecasting, go long, liquidate, go short, cover, the four profitable scenarios.

CORE
Never cut

Markets run on three instincts. To make a profit, something has to change. Price. The buy low, sell high principle. The rule is easy, obeying it is not. The price-time chart. Technical analysis does exactly two jobs. Identification. Forecasting. The four verb slides. The four ways a trade actually makes money. Quiz 1 opens with the four scenarios and its second item is the dual function, so none of this moves. Homework 1 is a charting exercise, which is why the price-time chart survives even at 75 minutes.

REINFORCEMENT
Cut when short

What identification hands you before you risk a peso, s20, which is the Identification slide told a second time. Figure 1.1, s28, which redraws the four scenarios.

ENRICHMENT
Cut first

What the survival instinct contains, s7.

FOLD
Say it, do not show it

When you cut the survival instinct slide, still say the herd line in one sentence on the instincts slide (s6), because Part 5 closes that loop on the real-world discounting slide, s162. At Short, teach the four verbs off the board square instead of the four term slides, s21 to s24.

s6

Open on the instincts. It reads like an odd place to begin a finance subject and that is the point: **markets are a survival behavior before they are a financial one.** Ask which of the three they would give up last, take two answers, move.

s7

Point at the herd line and tell them it comes back in Part 5. It does, on the real-world discounting slide s162, and closing that loop later is worth more than the thirty seconds it costs here.

s8 to s10

The move from survival to profit is the spine of the part. You cannot profit from something that never moves, so you need a variable of change, and price is the most convenient one we have. Then the mechanical rule, which sounds obvious.

s11, reveal s12

Check 1 Instincts, price, and the basic rule **CA**

First check of the session. Say out loud that the checks carry no marks before you read Q1, or the room will freeze.

s13

Then the trap. **To buy low you must first know today's price is low. That is a claim about the future.** Everything else this semester is an attempt to answer it.

s14

Homework 1 is a charting exercise on live charts with trendlines drawn and interpreted, so this slide is Core at every plan length including 75 minutes.

s15 to s17

The dual function is Quiz 1 item 2. Identification first, then forecasting, in that order, and say that the second depends on the first.

s18, reveal s19

Check 2 The chart and the two jobs **AC**

s21 to s24

WRITE THIS AND LEAVE IT UP FOR THE REST OF THE SESSION

	OPEN	CLOSE
BUY	go long	cover
SELL	go short	liquidate

The confusion is never with long. **It is that buying is not always bullish, because covering is a buy that ends a bearish position.** Say that sentence explicitly. Leaving the square up gives the room something to copy from at their own pace, and Part 6 points back at it.

s25, reveal s26

Check 3 The four words for a trade **AB**

s27

TRAP **Buying high and selling higher counts**

Warn them explicitly. **The department's Quiz 1 opens with exactly that item** and students who skim lose the mark.

s28

Figure 1.1, the mechanics of profiting from a change. It redraws the four scenarios, so it is the first thing to drop in this part and it goes at Long.

s29

Recap, then move. This is a clean stop if you need one, but Part 2 is the most examined part in the chapter, so do not spend the buffer here.

Three ways to forecast a price

Progress marker on screen reads `Part 2 of 6 - Forecasting`, 42 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
33 min	24 min	22 min	15 min

CHECKS	4 BA s38 5 BC s44 6 AD s50 7 DC s55 8 CA s63 9 DC s69
FIGURES	1.2 s32 cut at Long 1.3 s36 cut at Long 1.4 s37 cut at Long
OPEN	Three approaches, one question. Not three rivals: three answers to the same question.
CLOSE	The fundamentalist is cause, company, value. The technical analyst is effect, price, timing. Most professionals use both.
TERMS	Fundamental analysis, intrinsic value, undervalued and overvalued, the ratios, top-down, bottom-up, CAPM in passing, information analysis, non-public material information, technical analysis, the six famous definitions, technically and fundamentally based market timing, the fundamentalist profile, the technical analyst profile, the six streams of market action, OHLC.

CORE
Never cut

Three approaches, one question. Fundamental analysis. Intrinsic value. Undervalued and overvalued. Where fundamental analysis runs out of road. Information analysis. Why public information is usually late. Technical analysis. Murphy's first definition. Edwards and Magee on the market itself and not the goods. Pring on probabilities, never certainties. Murphy's second quote, price leads the known fundamentals. What technically based timing gives you. What fundamentally based timing gives you. The fundamentalist, in four lines. The technical analyst, in four lines. The six streams of market action. OHLC. **Quiz 1 spends items 3, 4, 5, 6 and 7 on this material, more than any other part gets**, which is why Part 2 loses the least between Long and Standard of any part in the chapter.

REINFORCEMENT
Cut when short

Figure 1.2, s32. Figures 1.3 and 1.4, s36 and s37. The top-down approach and the bottom-up approach, s41 and s42. How a fundamentalist turns accounts into a number, s40. Pring on the art of catching a reversal early, s54.

ENRICHMENT
Cut first

Edwards and Magee on recorded history and the word probable, and Pring on people continuing to make the same mistakes, which Part 5 says again on the behavior repeats slide s168. What technically based timing gives you, continued, s61.

FOLD
Say it, do not show it

Cutting the ratios slide is fine, losing its two sentences is not: **none of them has a time axis**, and **CAPM balances risk against expected return over a risk-free rate**. Quiz 1 item 3 carries a CAPM line. Name top-down and bottom-up as a pair in one sentence or check 5 loses its first question. Say the Pring art takeaway aloud or check 7 loses its second.

Whole part

> **RULE** **This is the longest part and the one with the most examinable content**
>
> Keep moving. Nothing here is safe to linger on.

s31

Frame the three approaches as **three answers to the same question** rather than as rivals. Fundamental analysis asks what the thing is worth, information analysis asks what the news says, technical analysis asks what the market is doing.

s34, s35

TRAP **A stock is overvalued when price is ABOVE intrinsic value**

Intrinsic value is the single most examinable term in this part, so make them write it down. Then the undervalued and overvalued pair, in that order, and say the trap out loud: the department has quizzed the reversed version.

s38, reveal s39

Check 4 Value and the three approaches **BA**

Written in the department's I, II, III, IV combination style. Point that out the first time it appears.

s40

On the ratios slide, name them and move. They belong to financial statement analysis, not to this subject. **The line that matters is that none of them has a time axis.**

s44, reveal s45

Check 5 Approaches and their limits **BC**

Q1 asks for top-down, so the fold sentence has to be said even when the two term slides come out. Cut at Short.

s46 to s47

Information analysis is short. The point is the lateness: by the time it is public it is usually in the price. That sets up Murphy's second quote later in this part.

s48 to s60

The six definitions

Six consecutive quote slides. Do not rush them, do not read them twice. Each has a takeaway line in green underneath and that is the sentence to say aloud.

- **Murphy.** Charts are the main tool, and the target is future price trends. This is the definition to give if an exam asks for one.
- **Edwards and Magee.** Recorded history, and the word that matters is **probable**.
- **Edwards and Magee.** The market itself, not the goods the market deals in. Cleanest one-line separation from fundamental analysis there is.
- **Pring.** It is an art, and the job is to catch a reversal early and stay until the evidence turns.
- **Pring.** Probabilities, never certainties. **If they remember one sentence from today, make it this one.**
- **Pring.** People continue to make the same mistakes. This is the behavioral reason the whole subject works.
- **Murphy again.** Market price leads the known fundamentals. This is the bridge into Part 5.

s50, reveal s51

Check 6 Information, and the first definition **AD**

s55, reveal s56

Check 7 Three definitions in a row **DC**

Q2 asks for the Pring art takeaway, so say it aloud even if the quote slide is cut. Cut at Short.

`s60 to s62` Make them notice for themselves that every fundamental line ends with "no precise price or time". **Do not point it out first.** Ask what the three lines have in common.

`s63, reveal s64`

Check 8 Timing, and who answers which question **CA**

Combination style again.

`s65 to s66` Close on the two profiles. Say the three-word summaries side by side: **the fundamentalist is cause, company, value; the technical analyst is effect, price, timing.** Then be fair and say most professionals use both.

`s67`

> **EXAMINED** **Flow of funds includes margin debt**
>
> That has been a past quiz item, almost verbatim. It is Quiz 1 item 6.

`s69, reveal s70`

Check 9 Market action and its data **DC**

`s71` Recap, then stop. This is the cleanest break point in the chapter: Parts 1 and 2 are the whole of the fundamental against technical argument, and Part 3 starts a new one.

Classifying technical analysis

Progress marker on screen reads `Part 3 of 6 - Classifications`, 27 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
28 min	**23** min	**15** min	**12** min

CHECKS	**10** **AA** s79 **11** **BD** s86 **12** **DC** s96
FIGURES	**1.6** s74 cut at Long **1.7** s82 cut at Long **1.8** s95 cut at Long
OPEN	The four branches are a filing system. Every tool in the whole book lands in exactly one of them.
CLOSE	The crowd destroys the edge it created, and once everyone abandons it, the edge comes back. Part 5 answers the other two objections.
TERMS	Classical, statistical, sentiment and behavioral analysis, the mean reverting or contrarian approach, the non-mean reverting or momentum approach, limit and stop entry orders, Darvas Box and Donchian channel breakouts, the advantages, the disadvantages, the self-fulfilling prophecy, the six-stage cycle, preempting.

CORE Never cut	Four branches, one subject, and all four branch terms, because booklet objective 2 is naming the forms of chart analysis. Two opposite bets about price. The mean reverting or contrarian approach. The non-mean reverting or momentum approach. Limit and stop entry orders. The Darvas Box line, said out loud. What technical analysis is genuinely good at. The honest disadvantages. The three big objections. The self-fulfilling prophecy, and both cycle slides. Quiz 1 item 8 is momentum and Darvas Box and contrarians, item 9 is the mean reverting tool set and names limit entry orders, and item 10 is random walk, which the room meets on the three objections slide. Objectives 5 and 7 both live here.
REINFORCEMENT Cut when short	Figure 1.6, s74. Figure 1.7, s82. Figure 1.8, s95. What technical analysis is genuinely good at, continued, s89.
ENRICHMENT Cut first	None. **Part 3 is the most heavily examined part per minute in the chapter.** What you save here you save on pace, not on slides.
FOLD Say it, do not show it	The advantages continued slide folds into the previous one, but keep its third line, because that line is what names the self-fulfilling prophecy before the term slide s92 arrives. At Short, run advantages and disadvantages as a single pass and skip the show of hands on the two bets.

s73 to s78

The four branches are a filing system. Every tool in the whole book lands in exactly one of them, and students should be able to sort a named technique into a branch on demand. **That is the examinable skill here, not the contents of any branch.**

The quickest way to fix the branches is by contrast. Classical is qualitative and visual, statistical is quantitative. Sentiment studies the mood, behavioral studies the participant.

s79, reveal s80

Check 10 The four branches **AA**

Booklet objective 2 in one check.

s81

Ask for a show of hands on which feels more natural **before** you teach either. Then say neither is right, they are right at different times, and that this is the whole difficulty of the subject.

s84

EXAMINED **Name the Darvas Box out loud**

The department has quizzed it as a momentum example. Quiz 1 item 8.

s85

The order type gives the trader away. Contrarians want a good price so they use limit orders; momentum traders want confirmation so they use stop orders. Ask which a contrarian prefers before you reveal it.

s86, reveal s87

Check 11 The two approaches **BD**

Combination style.

s88 to s91

RULE **Be honest about the disadvantages**

A course that only sells the method is a bad course. Three big objections get named on one slide: random walk, the strong form of EMH, and the self-fulfilling prophecy. **Say clearly that only the third is taught here and that Part 5 answers the other two properly**, so nobody spends Part 3 worrying about a term they have not met.

s92 to s94

The cycle runs over two slides, stages one to three then four to six. Stages one to three are the advantage, which is when the signal is worth trading. **Ask what happens when a strategy becomes too popular, then show stage four.** The crowd destroys the edge it created, and then, once everyone abandons it, the edge comes back. The process repeats.

s96, reveal s97

Check 12 Strengths, weaknesses, and the prophecy **DC**

s98

Recap, then move. Say before you leave the part that Part 5 answers random walk and the strong form of EMH properly, so the two loose objections do not sit there unanswered.

Subjectivity

Progress marker on screen reads `Part 4 of 6 - Subjectivity`, 39 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
42 min	34 min	20 min	15 min

CHECKS	13 BD s111 14 BA s116 15 DB s124 16 CA s134
FIGURES	1.9 s104 keep, all plans 1.10 s105 keep, all plans 1.11 s106 keep, all plans 1.12 s107 keep, all plans 1.13 s108 keep, all plans 1.14 s109 keep, all plans 1.15 s110 keep, all plans 1.16 s119 cut at Long 1.17 s121 keep to Standard 1.18 s122 cut at Long 1.19 s127 cut at Long 1.20 s129 keep to Standard 1.21 s133 cut at Long
OPEN	The chart is objective, the reading is subjective. The data is not the problem, the reader is.
CLOSE	Subjectivity is not a technical analysis problem, it is an analysis problem. Set the two-student trendline exercise before they leave.
TERMS	Objective and subjective aspects, the three activities, subjectivity, the twofold problem, contradictory, confirmatory and complementary signals, why indicators disagree, tick volume, the pattern size rule, interpretational and inferential subjectivity, selective perception, subjective objectivity, price, time and algorithmic filters, practice.

CORE
Never cut

Objective and subjective at the same time. Analysis is three separate activities. Subjectivity. The problem is twofold. **Figures 1.9 through 1.15, the whole sequence, in every plan including the 75 minute one.** Contradictory, confirmatory and complementary signals. Why indicators disagree in the first place. Resolving conflicting chart patterns, with Figure 1.17. The same fact, read two ways. Selective perception. Even the entry point is subjective, with Figure 1.20. Subjectivity shrinks with practice.

REINFORCEMENT
Cut when short

Figure 1.16, s119. Figure 1.18, s122. Figure 1.19, s127. Figure 1.21, s133. The price, time and algorithmic filters as three separate term slides, s130 to s132.

ENRICHMENT
Cut first

None. Nothing in this part is decorative.

FOLD
Say it, do not show it

Teach the three filters together off one slide in a minute rather than losing them, because check 16 asks for the time filter. At Short the sequence still runs exactly as written, at full speed, with the closing question; everything around it goes to one pass and the oil example gets 60 seconds.

`Whole part`

EXAMINED **Quiz 1 examines nothing from this part. Ignore that.**

Booklet objective 4 is this part by name, review questions 3 and 4 are answered here, and Homework 1 asks for trendlines drawn on a live chart and interpreted, which is Figure 1.20's argument and the closing exercise. **Part 4 is the last thing you cut, not the first.**

`s100 to s103`

This part makes students uncomfortable and it is supposed to. Say the frame early and repeat it: **the chart is objective, the reading is subjective. The data is not the problem, the reader is.**

Most students assume only the inference step is subjective. Step one already is, because you chose which two troughs to connect. Make them repeat the three activities back: identifying, interpreting, inferring.

s104 to s110

RULE **The seven readings are one move, not seven slides**

Same price chart every time: bare, then trendlines, moving averages, chart patterns, regression with divergence, regression with volume, and volatility bands with volume and MACD. **Run them straight through without stopping to teach any single one**, naming each reading in a few words as it comes up.

When the seventh is on screen, ask the room what changed between the first slide and this one. The answer is nothing except the analyst, and that is the argument of the whole part. It only lands if they have watched all seven go past in a row, so do not summarise the sequence, do not skip ahead, and **do not take questions in the middle of it.**

s111, reveal s112

Check 13 Where subjectivity lives **BD**

Straight after the sequence. Let the discomfort sit before you resolve it.

s113 to s115

Complementary signals is the most useful idea in this part. Walk the example slowly. A 20 period reading says slightly overbought and a 100 period reading says slightly oversold. That is not a contradiction, it is short term stretched and long term cheap, and the astute trader waits for cheap on both horizons and enters there.

s116, reveal s117

Check 14 Three kinds of signal **BA**

s120

TRAP **The pattern size rule is review question three**

Say so and make them write it down: **measure the patterns, and the larger formation's sentiment takes precedence, because larger formations speak for the longer term.** Give the caution too: an upside break of a large bearish formation can be violent precisely because it is unexpected, and short covering fuels it.

s123

The oil example is worth the time. Two fundamentalists, one fact, two opposite and defensible conclusions. Then land the caption: both of those analysts are fundamentalists. **Subjectivity is not a technical analysis problem, it is an analysis problem.** That is the fair-minded conclusion of the whole part and students should leave able to say it.

s124, reveal s125

Check 15 Conflicts, and how to settle them **DB**

s126

Selective perception is the most expensive habit in the room and every one of them will do it. Give the antidote plainly: **a signal that disagrees with you is the most informative thing on your screen.**

s128, s129

Subjective objectivity is review question four. Individually objective, collectively subjective. Use the program trading point to stop them believing that automation escapes the problem: the moment a parameter can be adjusted, subjectivity is back.

s134, reveal s135

Check 16 Bias, entries, and filters **CA**

Q2 asks for the time filter, so the three filters have to be named even when their term slides are folded. Cut at Short.

s136

Close on practice, and set the exercise before they leave: **two students, one chart, draw trendlines separately, then compare.** The point is to be comfortable with the difference, not to find a winner. This is Homework 1 in miniature.

s137

Recap. The sentence they should leave the part able to say is that subjectivity is an analysis problem, not a technical analysis problem. If that did not land, this is the recap to spend an extra thirty seconds on.

The assumptions underneath everything

Progress marker on screen reads `Part 5 of 6 - Assumptions`, 47 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
38 min	26 min	22 min	16 min

CHECKS	17 DB s145 18 BD s152 19 AC s157 20 AB s166 21 CD s174 22 DC s181
FIGURES	1.28 s143 keep to Standard 1.29 s144 cut at Long 1.27 s149 cut at Long 1.30 s160 cut at Long 1.31 s163 cut at Long 1.32 s164 cut at Long 1.25 s169 cut first 1.26 s170 cut first
OPEN	Write the three assumptions on the board and leave them there. Remove any one and technical analysis stops making sense.
CLOSE	An indicator that is badly designed will still become reliable if enough capital follows it. That closes the Part 3 loop.
TERMS	Market discounting, what markets can and cannot discount, what markets are really discounting, EMH, instantaneous and rational, the semi-efficient market, the weak, semi-strong and strong forms, random walk and the Markovian condition, real-world discounting, price versus value, behavior repeats, preempting and program trading, the market moves in trends, the challenges of trend following, the four applied assumptions, efficacy at various timeframes.

CORE Never cut	Three assumptions hold the subject up. Market discounting. What the market can and cannot discount. What the market is really discounting, with Figure 1.28. The Efficient Market Hypothesis. Efficient, under EMH, means two things. Why perfect efficiency cannot happen, with the handclap. The semi-efficient market. The weak, semi-strong and strong forms. Random walk. Random walk is not the same as EMH. What actually happens in the real world. Assumption two, behavior repeats. Three things that erode repeatability. Assumption three, the market moves in trends. Trend following is not free. Four assumptions you apply at the chart. **This part is booklet objective 1 on its own.** Quiz 1 item 10 is random walk, review questions 2, 5 and 6 all live here, and review question 1 needs the erosion slide for the shocks and algorithmic trading half of its answer.
REINFORCEMENT Cut when short	Figure 1.27, s149. Figure 1.29, s144. Figure 1.30, s160. Figures 1.31 and 1.32, s163 and s164. Price is not the same as value, s165. The four applied assumption term slides, s177 to s180, which restate the four numbered lines on the summary slide one at a time.
ENRICHMENT Cut first	Figures 1.25 and 1.26, s169 and s170, the angular symmetries and the ordered structure of price. Where technical analysis works best, s183. **These two are the first things to leave the chapter.**
FOLD Say it, do not show it	Teach the applied assumptions as a set off the summary slide s176 and you lose four slides and no content, and check 22 still works. At Short, teach the three forms of EMH off the ladder table below rather than the three term slides, and write the table on the board once. **Keep the handclap in every plan; it costs ten seconds.**

s139

WRITE THIS FIRST AND LEAVE IT UP

```
1.  The market discounts everything.
2.  Market behavior repeats itself.
3.  The market moves in trends.
```

The heaviest part of the chapter. **Remove any one and technical analysis stops making sense.** Leaving all three on the board gives the room a fixed reference for the next 38 minutes.

s141

TRAP **The two CANNOT lines are where the exam question lives**

Read them twice. Then explain the insider point carefully: insider information is non-public, but insider **trading** is visible in the market, so the market can discount it.

s142, s143

The five-line slide should be read at pace. The effect is cumulative. Then the payoff, which is the caption: **this is why a company can beat expectations and still fall.**

s145, reveal s146

Check 17 Discounting **DB**

s147 to s148

EXAMINED **EMH against market discounting is review question two**

Give the answer explicitly, dictate it if you have to: EMH requires that every participant reacts instantaneously and rationally. Technical analysis requires no such thing. **It requires only that the market discounts everything that becomes known to it, timely or untimely, rational or irrational.**

s150

RULE **Do the handclap test. Every plan. Ten seconds.**

Ask the whole room to clap the instant you say now. You will not get one sound. **If a hundred students in one room cannot coordinate, a million traders across a market certainly cannot.** That is the argument against perfect efficiency, delivered in a form they will remember.

s152, reveal s153

Check 18 EMH against market discounting **BD**

s154 to s156

THE THREE FORMS ARE THE HIGHEST VALUE ITEM IN THIS PART. MAKE THEM WRITE THE LADDER IN ORDER.

FORM	COVERS	KILLS
Weak	Past price information	Technical analysis
Semi-strong	All public information	Fundamental analysis as well
Strong	Public and private information	All analysis and forecasting

At Short, teach the three forms off this table and write it on the board once, instead of running the three term slides.

s157, reveal s158

Check 19 The three forms **AC**

Combination style.

s159 to s161

TRAP **Random walk is review question five, and the answer is no**

Draw the distinction in one sentence: **EMH says the market is too fast to beat, random walk says the market is meaningless.** Then the argument against it: watch how precisely price tests and reacts at a round number or an old high. Chance does not aim.

s162

This is where you close the herd loop from Part 1. Insiders accumulate, the public joins, the crowd overreacts, the insiders sell into it, a top forms. That is why a chart looks nothing like a coin flip.

s166, reveal s167

Check 20 Random walk and real markets **AB**

s168 to s173

On trend following, **emphasize the low winning percentage.** Students assume a good method wins most of its trades. It does not, and the drawdowns are the price of the profits.

The erosion slide s171 carries the shocks and algorithmic trading half of review question one, so it is Core even though it looks like a side note.

s174, reveal s175

Check 21 Repetition and trends **CD**

Cut at Short.

s176

The four applied assumptions are examinable as a set. Number them on the board. The fourth one is the self-fulfilling prophecy in formal dress, so close that loop from Part 3 and deliver the uncomfortable corollary: **an indicator that is badly designed will still become reliable if enough capital follows it.**

s181, reveal s182

Check 22 The four applied assumptions **DC**

Cut at Short.

s184

Recap. The three assumptions are still on the board from the start of the part, so read the recap against them, and rub the board only once Part 6 is under way.

Who is in the market

Progress marker on screen reads `Part 6 of 6 - Participants`, 22 slides including the divider and the recap.

FULL	LONG	STANDARD	SHORT
28 min	21 min	13 min	7 min

CHECKS	23 AC s189 24 AB s194 25 BC s204
FIGURES	1.33 s197 cut at Long 1.34 s199 cut at Long 1.35 s201 cut at Long
OPEN	A trade requires two people who disagree about price and agree on a number. Every price on every chart is that disagreement.
CLOSE	One underlying, five instruments, five different risk profiles, and only the first one needs a vault.
TERMS	The eight categories, retail and institutional, speculator and investor, supply side and demand side, professional and novice, discretionary and nondiscretionary, participants by time in the market, participants by method, the five main markets, derivatives, the five ways to participate in gold.

CORE
Never cut

The four contrast pair slides that carry the eight categories: retail and institutional, speculators and investors, supply side and demand side, professionals and novices. Discretionary and nondiscretionary traders. Sorted by time spent in the market. Sorted by method. The main markets, and Derivative, which hangs off it and which check 25 asks about. **Booklet objective 6 is categorizing participants by style and by time in markets, and objective 7 closes on the method ladder.** This is the part the old "Parts 1 to 3 are a complete unit" escape hatch threw away, and it is a stated objective, so it does not get thrown away again.

REINFORCEMENT
Cut when short

Figure 1.33, s197. Figure 1.34, s199. Figure 1.35, s201. The cast of a market, s186. Five ways to own gold, s203.

ENRICHMENT
Cut first

None.

FOLD
Say it, do not show it

Fold gold into the Derivative term slide s202 in one sentence, **a gold backed exchange traded fund is a derivative**, because check 25 asks for exactly that. The two ladders stay on screen in every plan down to Standard; at Short they are the only two slides in this part that get their full minute.

Whole part

> **EXAMINED** **Quiz 1 examines nothing from this part either. The booklet does, twice.**
>
> At Short this is the one objective that degrades, and the compensation is on the run plans page. The lightest part, and a good one to end on.

s186

Open by saying **a trade requires two people who disagree about price and agree on a number.** Every price on every chart is that disagreement.

s187 to s192

The eight categories are a list to be able to reproduce: retail, institutional, speculator, supply side, demand side, professional, investor, novice. The deck teaches them in four contrast pairs, which is how they stick.

Two of them are worth a sentence beyond the definition

- **Supply side and demand side cuts across retail and institutional.** The supply side gets paid whether the demand side wins or loses.
- **Novice is a stage, not a verdict.** Everyone in the room is one today, and Part 4 already said that subjectivity in pattern recognition falls with practice.

s189, reveal s190

Check 23 Who is who **AC**

Kept in every plan, including Short.

s194, reveal s195

Check 24 Categories and methods **AB**

Cut at Short.

s196, s198

Both ladders are stated learning objectives in the course booklet, so make sure both slides get their minute. The method ladder closes the loop with Part 3: reversal traders are the contrarians, trend traders are the momentum side.

s198

TRAP **Give the scale trading warning plainly**

Averaging against price until it turns is the highest risk method on that list, and **it is how accounts end.**

s200 to s203

Finish on the gold slide. One underlying, five instruments, five different risk profiles, and only the first one needs a vault.

s204, reveal s205

Check 25 Markets and instruments **BC**

Last check of the session. Q2 is the time ladder, which is why this one survives at Short even though objective 6 degrades there.

s206 to s209

Recap, then the three closing slides. Do not let the session end on a check: the last thing they hear should be what Chapter 1 was for, and what is coming next.

Before they leave

s208

The chapter review questions

Short crib. The full answers are in the deck, in the parts named.

	QUESTION	WHERE	ANSWER IN ONE LINE
1	Challenges to technical analysis	Part 3	Subjectivity of interpretation, disruption of repeatability by shocks and by algorithmic trading, the difficulty of inference, random walk, the strong form of EMH, and the self-fulfilling prophecy.
2	Market discounting against EMH	Part 5	EMH requires instantaneous and rational reaction by all participants. Technical analysis requires only that the market discounts what becomes known to it.
3	Resolving conflicting signals or patterns	Part 4	Check the time horizons first, since apparently conflicting signals are often complementary. For chart patterns, measure them: the larger formation's sentiment takes precedence until its own level is breached.
4	Why identifying a trend change is subjective	Part 4	Each act of identification is objective, but you chose the trendline, and an alternative trendline gives a different answer. Individually objective, collectively subjective.
5	Is random walk a true reflection of the markets?	Part 5	No. Markets are driven by perception and expectation, and participants react in predictable ways at psychologically significant prices.
6	The three levels of discounting under EMH	Part 5	Weak, semi-strong and strong. The ladder table is on the Part 5 page.
7	A good definition of technical analysis	Part 2	Murphy: the study of market action, primarily through the use of charts, for the purpose of forecasting future price trends. Slide 49.
8	Advantages and disadvantages	Part 3	Four slides, s88 to s91.

s207 to s209

Quiz preparation to mention

The last three slides.

Say this

RULE **Quiz 1 covers Chapters 1 and 2 together**

Many department items use the "I, II, III, IV, which combination is correct" style, so five of the twenty five checks are written that way: **checks 4, 8, 11, 19 and 25.** Tell them to expect it, and that the trick is to find the one false statement rather than to verify all four.

Homework 1

Homework 1 is a charting exercise, not multiple choice. It asks for a live chart with trendlines drawn and the trend interpreted, which is exactly the Part 4 closing exercise. **Setting that exercise today is preparation for it.**

s209

Close by naming what comes next. Dow Theory is Chapter 2 and the other half of Quiz 1, so the preview is not a courtesy.